

Bayesian Linear Regression and Variable Selection

Horseshoe Linear Regression

The model, the Gibbs sampler,
and variable selection

Detailed research notes

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Purpose. Develop Gaussian linear regression with a horseshoe prior from the likelihood to an implementable Gibbs sampler. Every full conditional is derived, the distribution conventions are fixed explicitly, and posterior shrinkage is separated from the decision to select variables.

Main convention. The coefficient prior is scaled by the noise variance: $\beta_j \mid \sigma^2, \tau^2, \lambda_j^2 \sim \mathcal{N}(0, \sigma^2 \tau^2 \lambda_j^2)$. The intercept is separate and unpenalized. The main derivation uses a proper inverse-gamma prior on σ^2 , with common alternatives explained later.

Scope. These are mathematical and implementation notes, not an empirical comparison. Worked quantities are analytic illustrations. No simulated data, fitted model, convergence result, or benchmark is reported.

Editable source: `horseshoe_linear_regression.tex`

The bibliography links to the original methodological sources.

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1 The question the horseshoe answers

Suppose a response may depend on many predictors, but we expect only a small number of substantial effects. Ordinary least squares estimates each coefficient without using that expectation. When predictors are numerous or correlated, the estimates can be unstable; when $p \geq n$, an unregularized coefficient vector may not even be identifiable.

The horseshoe expresses sparsity through a continuous prior. A shared *global* scale encourages coefficients to be small, while a separate *local* scale lets each coefficient escape strong shrinkage when supported by the likelihood. This global–local construction is the central idea of Carvalho, Polson, and Scott [1, 2].

There are three different tasks:

1. **Estimation:** describe plausible values of each coefficient and of the regression function.
2. **Prediction:** describe responses at new predictor values, including both parameter uncertainty and observation noise.
3. **Selection:** decide which variables to report or retain, using a stated rule or loss function.

A horseshoe posterior directly addresses the first two. The third requires an additional decision: the ordinary horseshoe does not assign positive probability to any coefficient being exactly zero.

1.1 Reading guide

Sections 2–3 explain the model and its shrinkage behavior. Sections 4–7 give the augmented posterior and derive the complete Gibbs sweep. Sections 8 and 9 address implementation and prior calibration. Sections 10–12 cover inference, selection, and verification. The appendices distinguish alternative parameterizations and provide a compact reference sheet.

1.2 Notation and distribution conventions

Vectors are columns. In particular, $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, and $\beta \in \mathbb{R}^p$. The i th row of X is $x_i^\top$, and X_j denotes its j th column. The intercept is $\alpha \in \mathbb{R}$. The symbol $|\cdot$ means conditioning on the data, fixed hyperparameters, and all current unknowns except the variable being updated.

Symbol	Meaning
n, p	Number of observations and number of penalized slopes; the intercept is not counted in p .
σ^2	Observation-noise variance; $\sigma = \sqrt{\sigma^2} > 0$.
τ, s_τ	Global shrinkage scale and the fixed scale of its half-Cauchy prior.
λ_j	Local shrinkage scale for slope j .
ν_j, ξ	Positive auxiliary variables used to represent half-Cauchy priors.
a_σ, b_σ	Shape and inverse-gamma scale for the noise-variance prior.
D, A, m	Prior variance multiplier, conditional precision multiplier, and conditional coefficient mean.

We use $\mathcal{N}_k(m, V)$ with *covariance* V , not precision. The inverse-gamma convention is

$$Z \sim \text{IG}(a, b) \iff p(z) = \frac{b^a}{\Gamma(a)} z^{-a-1} \exp(-b/z), \quad z > 0. \quad (1.1)$$

Thus $1/Z \sim \text{Gamma}(a, \text{rate} = b)$. If software uses gamma shape and scale, generate $G \sim \text{Gamma}(a, \text{scale} = 1/b)$ and return $1/G$. The mean $b/(a-1)$ exists only when $a > 1$; the mode is $b/(a+1)$.

A half-Cauchy distribution here is parameterized by its *scale* s :

$$T \sim \text{C}^+(0, s), \quad p(t) = \frac{2s}{\pi(s^2 + t^2)}, \quad t > 0. \quad (1.2)$$

The symbols τ and λ_j are standard-deviation multipliers; τ^2 and λ_j^2 are the variables sampled in the augmented Gibbs scheme. Confusing a scale with its square changes the model.

2 The regression model and the prior

2.1 Likelihood

Assume conditionally independent, homoscedastic Gaussian errors:

$$y_i = \alpha + x_i^\top \beta + \varepsilon_i, \quad \varepsilon_i \mid \sigma^2 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (2.1)$$

$$y \mid \alpha, \beta, \sigma^2 \sim \mathcal{N}_n(\alpha \mathbf{1}_n + X\beta, \sigma^2 I_n). \quad (2.2)$$

Consequently,

$$p(y \mid \alpha, \beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{\|y - \alpha \mathbf{1}_n - X\beta\|^2}{2\sigma^2} \right\}. \quad (2.3)$$

This likelihood is an assumption about the errors, not a requirement that predictors themselves be normally distributed.

2.2 The noise-scaled horseshoe hierarchy

Our main model is

$$\beta_j \mid \sigma^2, \tau^2, \lambda_j^2 \stackrel{\text{ind}}{\sim} \mathcal{N}(0, \sigma^2 \tau^2 \lambda_j^2), \quad j = 1, \dots, p, \quad (2.4)$$

$$\lambda_j \stackrel{\text{iid}}{\sim} \text{C}^+(0, 1), \quad \tau \sim \text{C}^+(0, s_\tau), \quad (2.5)$$

$$\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma), \quad p(\alpha) \propto 1, \quad (2.6)$$

where $a_\sigma, b_\sigma, s_\tau > 0$. The scale priors and the noise-variance prior are mutually independent. The slopes are independent *conditional* on their scales and σ^2 ; they are not generally independent after integrating out the shared scales.

Define

$$D = \tau^2 \text{diag}(\lambda_1^2, \dots, \lambda_p^2), \quad \beta \mid \sigma^2, D \sim \mathcal{N}_p(0, \sigma^2 D). \quad (2.7)$$

The prior precision for coefficient j is $1/(\sigma^2 \tau^2 \lambda_j^2)$. Small τ encourages shrinkage throughout the model. A large λ_j weakens the penalty for its own coefficient. A small λ_j strengthens that penalty.

The conventional unit global scale is $s_\tau = 1$. It is a transparent reference choice, not a universal calibration. Section 9 explains how sample size, predictor scaling, and an anticipated degree of sparsity affect a more informed choice.

A notation boundary for this project

The λ_j 's in these notes are horseshoe local scales with half-Cauchy priors. They are not the lambda parameter of a quantile-regression sampler. The project's $\text{Gamma}(1, \text{rate} = \sqrt{n})$ convention for that different parameter does not replace the horseshoe hierarchy.

2.3 Centering, scaling, and the intercept

Predictor units affect the meaning of a prior on coefficients. Unless units carry a reason for unequal prior scales, center each nonconstant predictor and divide it by a recorded scale. One precise convention is

$$\bar{x}_j = \frac{1}{n} \sum_i x_{ij}^{\text{raw}}, \quad s_j^2 = \frac{1}{n} \sum_i (x_{ij}^{\text{raw}} - \bar{x}_j)^2, \quad x_{ij} = \frac{x_{ij}^{\text{raw}} - \bar{x}_j}{s_j}. \quad (2.8)$$

Then $\mathbf{1}_n^\top X = 0$ and $X_j^\top X_j = n$. Using sample standard deviation instead gives $X_j^\top X_j = n - 1$; either is valid if used consistently. Constant columns must be removed or handled separately.

An intercept is an overall baseline, not a candidate sparse predictor. We therefore keep it outside D . Under centered predictors, $\alpha \mid \cdot \sim \mathcal{N}(\bar{y}, \sigma^2/n)$. Keeping this draw retains baseline uncertainty for prediction.

Centering y and setting its intercept to zero is not automatically equivalent to integrating out an unknown intercept. With a flat intercept prior, exact integration reduces the residual dimension from n to $n - 1$. Appendix A gives the resulting formulas. The main sampler avoids ambiguity by sampling α explicitly.

For validation or prediction, estimate preprocessing constants using training data only, and apply those same constants to held-out observations. If the response was also transformed as $y' = (y^{\text{raw}} - c_y)/s_y$, then

$$\beta_j^{\text{raw}} = \frac{s_y}{s_j} \beta_j, \quad \alpha^{\text{raw}} = c_y + s_y \alpha - \sum_j \frac{s_y \beta_j \bar{x}_j}{s_j}, \quad \sigma^{\text{raw}} = s_y \sigma. \quad (2.9)$$

Apply these transformations to every posterior draw, so uncertainty transforms with the parameters.

2.4 What makes this prior a horseshoe?

Conditional normality makes computation tractable. Mixing that normal over a half-Cauchy local scale gives a marginal coefficient density with an unbounded spike at zero and heavy tails. For fixed positive τ, σ , it remains a proper continuous density. An infinite density value at one point is not a positive probability at that point.

The spike supports estimates close to zero; the tails accommodate large effects without a Gaussian prior's strong tail penalty. This does not guarantee that a weak signal is retained or that a particular correlated predictor is identified. The likelihood and the global-scale prior still matter.

3 Understanding shrinkage through an orthogonal design

3.1 A one-coordinate normal calculation

For intuition, suppose the predictor columns are mutually orthogonal and orthogonal to the intercept. Let $c_j = X_j^\top X_j > 0$. The least-squares coordinate is

$$\hat{\beta}_j = \frac{X_j^\top (y - \alpha \mathbf{1}_n)}{c_j}, \quad \hat{\beta}_j \mid \beta_j, \sigma^2 \sim \mathcal{N}\left(\beta_j, \frac{\sigma^2}{c_j}\right).$$

Combining this normal likelihood with (2.4) adds their precisions. Define the shrinkage factor

$$\kappa_j = \frac{1}{1 + c_j \tau^2 \lambda_j^2}. \quad (3.1)$$

The conditional posterior becomes

$$\beta_j \mid \cdot \sim \mathcal{N}\left((1 - \kappa_j) \hat{\beta}_j, \frac{\sigma^2}{c_j} (1 - \kappa_j)\right). \quad (3.2)$$

Thus κ_j near one means strong shrinkage; κ_j near zero means weak shrinkage. The noise variance cancels out of κ_j because the likelihood and coefficient prior share the factor σ^2 .

For a unit-length column, $c_j = 1$. Under (2.8), $c_j = n$. Using $1/(1 + \tau^2 \lambda_j^2)$ indiscriminately in standardized regression misses this factor of n .

3.2 Why the name? A transformation of the local scale

For a unit-information coordinate with $c_j = 1$ and fixed $\tau = 1$, $\lambda_j \sim C^+(0, 1)$ implies $\kappa_j = 1/(1 + \lambda_j^2) \sim \text{Beta}(1/2, 1/2)$. Its density is high near both endpoints. One end represents substantial shrinkage and the other little shrinkage.

More generally put $t = \sqrt{c_j} \tau$. Inverting $\kappa = 1/(1 + t^2 \lambda^2)$ and including the Jacobian gives

$$p(\kappa \mid \tau) = \frac{t}{\pi \sqrt{\kappa(1 - \kappa)} \{1 + (t^2 - 1)\kappa\}}, \quad 0 < \kappa < 1. \quad (3.3)$$

This is a *prior* density at fixed τ . Data change the posterior distribution of κ ; it is not a prior inclusion probability.

3.3 An analytic illustration

Take an orthogonal column with $c_j = 100$, and hold $\tau = 0.05$ fixed. Then $c_j \tau^2 = 0.25$. The following values are direct substitutions in (3.1); they are not fitted estimates or simulation results.

Local scale λ_j	$c_j \tau^2 \lambda_j^2$	κ_j	Fraction retained $1 - \kappa_j$
0.2	0.01	0.9901	0.0099
1	0.25	0.8000	0.2000
10	25	0.0385	0.9615

The same global scale can nearly eliminate one coordinate and preserve most of another. In actual inference the scales are random and must be averaged over, rather than fixed at convenient values.

3.4 The limitation with correlated predictors

For a general design, the conditional mean is $(X^\top X + D^{-1})^{-1} X^\top (y - \alpha \mathbf{1}_n)$. Off-diagonal entries of $X^\top X$ couple the coefficients. There is generally no exact formula that independently multiplies each ordinary least-squares coefficient by $1 - \kappa_j$.

In particular, two nearly interchangeable predictors can trade explanatory roles across posterior draws. Their joint contribution may be stable while their individual coefficients remain uncertain. Shrinkage factors remain useful scale summaries, but their coordinatewise interpretation is exact only in the orthogonal setting.

4 Turning half-Cauchy priors into conjugate updates

4.1 The auxiliary-variable identity

The Gibbs sampler uses the inverse-gamma representation employed by Makalic and Schmidt [3]. For any $s > 0$, introduce $V > 0$ and write

$$Z \mid V \sim \text{IG}(1/2, 1/V), \quad V \sim \text{IG}(1/2, 1/s^2). \quad (4.1)$$

Then $\sqrt{Z} \sim C^+(0, s)$.

To verify the identity, retain all factors that depend on z or v :

$$p(z, v) = \frac{1}{\pi s} z^{-3/2} v^{-2} \exp \left\{ -\frac{1/z + 1/s^2}{v} \right\}.$$

For $C > 0$, substitution $u = C/v$ gives $\int_0^\infty v^{-2} e^{-C/v} dv = 1/C$. Hence

$$p(z) = \frac{1}{\pi s} \frac{z^{-3/2}}{1/z + 1/s^2} = \frac{s}{\pi \sqrt{z}(s^2 + z)}. \quad (4.2)$$

Setting $z = t^2$, the Jacobian is $2t$, so $p(t) = 2s/\{\pi(s^2 + t^2)\}$, exactly (1.2).

The auxiliary variable has not approximated or replaced the half-Cauchy. Integrating it out recovers the original prior exactly. Its purpose is to make the conditional distributions easy to sample.

4.2 The augmented horseshoe model

Apply (4.1) independently to every local scale and once to the global scale:

$$\lambda_j^2 \mid \nu_j \sim \text{IG}(1/2, 1/\nu_j), \quad \nu_j \sim \text{IG}(1/2, 1), \quad (4.3)$$

$$\tau^2 \mid \xi \sim \text{IG}(1/2, 1/\xi), \quad \xi \sim \text{IG}(1/2, 1/s_\tau^2). \quad (4.4)$$

Together with the Gaussian likelihood, coefficient prior, and (2.6), this is the model sampled below. In this chosen parameterization, s_τ appears in the prior on ξ , and later in the full conditional for ξ . Moving it into a different update without redefining the augmentation would be an error.

The variables ν_j and ξ are computational devices. They are neither regression coefficients nor indicators that a predictor is included. They can be omitted from scientific summaries, but must be maintained in the chain state if these conditional updates are used.

5 The posterior kernel and a recipe for deriving conditionals

Let $r = y - \alpha \mathbf{1}_n - X\beta$. Up to constants independent of all unknowns, the augmented joint posterior density with respect to $d\alpha d\beta d\sigma^2 d\tau^2 d\lambda^2 d\nu d\xi$ is

$$\begin{aligned} \pi(\alpha, \beta, \sigma^2, \tau^2, \lambda^2, \nu, \xi \mid y, X) &\propto (\sigma^2)^{-n/2} \exp\left(-\frac{r^\top r}{2\sigma^2}\right) \\ &\times (\sigma^2)^{-p/2} (\tau^2)^{-p/2} \prod_{j=1}^p (\lambda_j^2)^{-1/2} \exp\left(-\sum_j \frac{\beta_j^2}{2\sigma^2 \tau^2 \lambda_j^2}\right) \\ &\times (\sigma^2)^{-a_\sigma-1} \exp(-b_\sigma/\sigma^2) \\ &\times \prod_{j=1}^p \left[\nu_j^{-1/2} (\lambda_j^2)^{-3/2} \exp\left(-\frac{1}{\nu_j \lambda_j^2}\right) \nu_j^{-3/2} \exp(-1/\nu_j) \right] \\ &\times \xi^{-1/2} (\tau^2)^{-3/2} \exp\left(-\frac{1}{\xi \tau^2}\right) \xi^{-3/2} \exp\left(-\frac{1}{s_\tau^2 \xi}\right). \end{aligned} \quad (5.1)$$

The flat intercept prior contributes no factor. Fixed factors such as $(s_\tau^2)^{-1/2}$ can be suppressed here.

To derive a full conditional, keep only factors involving the variable being updated, holding all other variables fixed. For inverse-gamma updates, identify the exponent of the variable and the coefficient of its reciprocal. For normal updates, complete a quadratic square.

For example, the factor $\nu_j^{-1/2}$ comes from the normalization of $\lambda_j^2 \mid \nu_j$. It cannot be discarded when updating ν_j . Likewise, $(\sigma^2)^{-p/2}$ in the coefficient prior is essential when updating σ^2 . Many incorrect samplers arise from treating a normalizing constant as constant in the wrong variable.

6 Derivation of every full conditional

6.1 Intercept: one unpenalized normal draw

Write $z = y - X\beta$ and $\bar{z} = n^{-1} \mathbf{1}_n^\top z$. Then

$$\sum_i (z_i - \alpha)^2 = \sum_i (z_i - \bar{z})^2 + n(\alpha - \bar{z})^2.$$

Only the last term involves α . With its flat prior,

$$\alpha \mid \cdot \sim \mathcal{N}\left(\frac{\mathbf{1}_n^\top (y - X\beta)}{n}, \frac{\sigma^2}{n}\right). \quad (6.1)$$

For centered X , the mean is simply $\bar{y}$. If instead $\alpha \sim \mathcal{N}(m_\alpha, v_\alpha)$, independent of σ^2 , replace the variance by $V_\alpha = (n/\sigma^2 + 1/v_\alpha)^{-1}$ and the mean by $V_\alpha \{\mathbf{1}_n^\top (y - X\beta)/\sigma^2 + m_\alpha/v_\alpha\}$. This independent proper intercept prior does not add a power of σ^2 .

6.2 Regression coefficients: a multivariate normal block

Set $\tilde{y} = y - \alpha \mathbf{1}_n$. The terms involving β are

$$\pi(\beta \mid \cdot) \propto \exp\left\{-\frac{1}{2\sigma^2} [(\tilde{y} - X\beta)^\top (\tilde{y} - X\beta) + \beta^\top D^{-1} \beta]\right\}.$$

Expand the quadratic and define

$$A = X^\top X + D^{-1}, \quad b = X^\top \tilde{y}, \quad m = A^{-1}b. \quad (6.2)$$

Because $Am = b$, completing the square gives

$$\beta^\top A \beta - 2\beta^\top b = (\beta - m)^\top A (\beta - m) - m^\top Am.$$

The final term is constant in β , and therefore

$$\boxed{\beta \mid \cdot \sim \mathcal{N}_p(m, \sigma^2 A^{-1}).} \quad (6.3)$$

For any nonzero v , $v^\top A v = \|Xv\|^2 + \sum_j v_j^2 / (\tau^2 \lambda_j^2) > 0$. Thus A is positive definite for finite positive scales, even if $X^\top X$ is singular. The conditional normal distribution is well defined for $p > n$. This is a statement about regularization, not a claim that the data identify every coefficient separately.

Drawing the entire vector as a block respects its conditional dependence. Replacing the draw by m would discard posterior variability and cease to be a Gibbs sampler. Section 8 describes how to draw this normal without computing a matrix inverse.

6.3 Noise variance: likelihood plus coefficient-prior contribution

Define the two nonnegative sums

$$R = \|y - \alpha \mathbf{1}_n - X\beta\|^2, \quad Q = \beta^\top D^{-1} \beta = \sum_{j=1}^p \frac{\beta_j^2}{\tau^2 \lambda_j^2}. \quad (6.4)$$

Combining the likelihood, coefficient-prior normalization, and noise prior,

$$\pi(\sigma^2 \mid \cdot) \propto (\sigma^2)^{-\{a_\sigma + (n+p)/2\}-1} \exp \left\{ -\frac{b_\sigma + (R+Q)/2}{\sigma^2} \right\}.$$

Comparison with (1.1) yields

$$\boxed{\sigma^2 \mid \cdot \sim \text{IG} \left(a_\sigma + \frac{n+p}{2}, b_\sigma + \frac{R+Q}{2} \right).} \quad (6.5)$$

The $p/2$ contribution is required because the prior variance of each of the p slopes contains σ^2 . The term Q is required for the same reason. The intercept contributes neither an extra $1/2$ nor an intercept penalty because its prior is flat. The likelihood still contributes $n/2$, since α is conditioned on here, not integrated out.

6.4 Local variance multiplier for each coefficient

For $\ell_j = \lambda_j^2$, the coefficient prior contributes $\ell_j^{-1/2} \exp\{-\beta_j^2/(2\sigma^2\tau^2\ell_j)\}$, and the augmented prior contributes $\ell_j^{-3/2} \exp\{-1/(\nu_j\ell_j)\}$. Hence

$$\pi(\ell_j \mid \cdot) \propto \ell_j^{-2} \exp \left\{ -\frac{\nu_j^{-1} + \beta_j^2/(2\sigma^2\tau^2)}{\ell_j} \right\},$$

which is

$$\lambda_j^2 \mid \cdot \sim \text{IG} \left(1, \frac{1}{\nu_j} + \frac{\beta_j^2}{2\sigma^2\tau^2} \right), \quad j = 1, \dots, p. \quad (6.6)$$

These updates are conditionally independent across j . A large $\beta_j^2/(\sigma^2\tau^2)$ increases the inverse-gamma scale and allows a larger local variance. This is how a coefficient can resist global shrinkage. It is a stochastic update, not a deterministic monotone rule.

The conditional shape is one. Its mean is infinite, so an argument based on the “conditional mean of λ_j^2 ” is not available. Simulating this proper distribution is nevertheless well defined. Scale medians, log-scales, and shrinkage factors are often more interpretable summaries.

6.5 Global variance multiplier

Put $t = \tau^2$. All p coefficient priors contribute a total factor $t^{-p/2}$, while the augmented global prior contributes $t^{-3/2}$. Thus

$$\pi(t \mid \cdot) \propto t^{-(p+3)/2} \exp \left\{ -\frac{\xi^{-1} + \sum_j \beta_j^2 / (2\sigma^2\lambda_j^2)}{t} \right\}.$$

Matching the power $-a - 1$ gives

$$\tau^2 \mid \cdot \sim \text{IG} \left(\frac{p+1}{2}, \frac{1}{\xi} + \frac{1}{2\sigma^2} \sum_{j=1}^p \frac{\beta_j^2}{\lambda_j^2} \right). \quad (6.7)$$

Unlike a local update, this update aggregates information from all slopes. Large local scales can accommodate a few substantial coefficients while allowing the global scale to remain small for the others.

6.6 Auxiliary variables

For ν_j , the normalization in $\lambda_j^2 \mid \nu_j$ supplies $\nu_j^{-1/2}$. Multiplying by its own prior gives

$$\pi(\nu_j \mid \cdot) \propto \nu_j^{-2} \exp \left\{ -\frac{1 + 1/\lambda_j^2}{\nu_j} \right\}.$$

Similarly,

$$\pi(\xi \mid \cdot) \propto \xi^{-2} \exp \left\{ -\frac{1/s_\tau^2 + 1/\tau^2}{\xi} \right\}.$$

Therefore,

$$\nu_j \mid \cdot \sim \text{IG} \left(1, 1 + \frac{1}{\lambda_j^2} \right), \quad j = 1, \dots, p, \quad (6.8)$$

$$\xi \mid \cdot \sim \text{IG} \left(1, \frac{1}{s_\tau^2} + \frac{1}{\tau^2} \right). \quad (6.9)$$

When $s_\tau = 1$, the last scale reduces to $1 + 1/\tau^2$. No Metropolis acceptance step and no slice-sampling subroutine is needed for any of these updates in the stated Gaussian model.

7 How the Gibbs sampler works

7.1 From conditional draws to a joint posterior sample

A Gibbs sampler repeatedly replaces one block of unknowns by a draw from its conditional posterior given the current remaining blocks. If the chain is already distributed according to the joint posterior, any one exact conditional update leaves that joint distribution unchanged. A sequence of such updates also leaves it unchanged.

This invariance identifies the target distribution; it does not say that a chain started arbitrarily is immediately at stationarity. Nor does it make successive draws independent. Appropriate regularity and adequate exploration are needed for the empirical distribution of a long run to approximate the posterior. Diagnostics assess finite-run behavior, not just the correctness of the formulas.

7.2 A complete sweep with explicit update order

Store $\{\alpha, \beta, \sigma^2, \lambda^2, \tau^2, \nu, \xi\}$. Choose finite initial values with every variance and auxiliary variable strictly positive. For example, start α near $\bar{y}$, β near zero, $\lambda_j^2 = \nu_j = 1$, a positive noise variance, $\tau^2 = s_\tau^2$, and $\xi = 1$. These are starting values, not prior draws or estimated hyperparameters. Use dispersed starts across chains.

At iteration $t = 1, \dots, T$, carry out:

1. **Update the intercept.** Draw $\alpha^{(t)}$ using (6.1) with $\beta^{(t-1)}, \sigma^{2(t-1)}$.
2. **Update all slopes together.** Form D from $\tau^{2(t-1)}, \lambda^{2(t-1)}$, set $\tilde{y} = y - \alpha^{(t)} \mathbf{1}_n$, and draw $\beta^{(t)}$ from (6.3) with $\sigma^{2(t-1)}$.
3. **Update the noise variance.** Evaluate R using the new $\alpha^{(t)}, \beta^{(t)}$, and Q using the still-current old scales. Draw $\sigma^{2(t)}$ from (6.5).
4. **Update local variances.** For every j , draw $\lambda_j^{2(t)}$ from (6.6) with $\beta_j^{(t)}, \sigma^{2(t)}, \tau^{2(t-1)}, \nu_j^{(t-1)}$. The local draws can be generated as one conditionally independent batch.
5. **Update the global variance.** Draw $\tau^{2(t)}$ from (6.7) using $\beta^{(t)}, \sigma^{2(t)}, \lambda^{2(t)}$ and $\xi^{(t-1)}$.
6. **Update the auxiliary variables.** Draw $\nu_j^{(t)}$ from (6.8) using $\lambda_j^{2(t)}$, and draw $\xi^{(t)}$ from (6.9) using $\tau^{2(t)}$.
7. **Save the completed state.** After the chosen initial transient, store draws and any desired predictive or shrinkage summaries. Save checkpoints separately if the run must be restartable.

Each update uses the most recently available values. Other sequential scan orders can be valid, but updating every block simultaneously using only the previous iteration's values is generally not the same Gibbs transition and need not preserve the posterior.

7.3 Why the updates adapt to sparsity

The coefficient draw combines data fit with the present penalties. The local updates then accommodate the magnitudes of individual coefficients. The global update adjusts the overall scale

after accounting for those local exceptions. Finally, the auxiliary updates keep the scale-mixture representation consistent with the intended half-Cauchy priors.

This feedback repeats throughout the run. One iteration is not a fitted answer; the posterior distribution is represented by the retained collection of complete states. A small coefficient at one iteration does not mean that the variable has been permanently removed.

7.4 An implementation skeleton

The following is language-neutral pseudocode. `IGdraw(a,b)` follows (1.1), and `NormalBlock` performs the random draw in (6.3), not just its conditional mean.

```

precompute XtX = transpose(X) * X, Xty = transpose(X) * y
precompute Xt1 = transpose(X) * ones(n)
initialize alpha, beta, sigma2, lambda2, tau2, nu, xi
for iteration = 1,...,T:
    alpha = Normal(mean(y - X * beta), sigma2 / n)
    d = tau2 * lambda2
    A = XtX + diag(1 / d)
    b = Xty - alpha * Xt1
    beta = NormalBlock(A, b, sigma2)
    residual = y - alpha * ones(n) - X * beta
    R = dot(residual, residual)
    Q = sum(beta^2 / d)
    sigma2 = IGdraw(a_sigma + (n+p)/2, b_sigma + (R+Q)/2)
    lambda2 = IGdraw(1, 1/nu + beta^2/(2*sigma2*tau2))
    tau2 = IGdraw((p+1)/2, 1/xi + sum(beta^2/lambda2)/(2*sigma2))
    nu = IGdraw(1, 1 + 1/lambda2)
    xi = IGdraw(1, 1/s_tau^2 + 1/tau2)
    save completed state if retained

```

Arithmetic on vectors is elementwise except for explicit matrix products. The second argument of `Normal` above is a variance. For $p \gg n$, the next section replaces the $p \times p$ block construction, so $X^\top X$ need not be stored.

8 Sampling the Gaussian block accurately and efficiently

8.1 Cholesky sampling when the coefficient dimension is manageable

Although (6.3) is written using A^{-1} , implementation should use linear solves. Let $A = LL^\top$, where L is lower triangular. First solve

$$Lq = b, \quad L^\top m = q.$$

Draw $z \sim \mathcal{N}_p(0, I_p)$ and solve $L^\top v = z$. Return

$$\beta = m + \sigma v. \tag{8.1}$$

Indeed,

$$\text{Cov}(v) = L^{-\top} I_p L^{-1} = (LL^\top)^{-1} = A^{-1}.$$

The transpose matters. Solving $Lv = z$ instead would generally give $L^{-1}L^{-\top}$, which is not A^{-1} . If a software package returns an upper-triangular factor U with $A = U^{\top}U$, solve $Uv = z$ instead.

For a dense design, precomputing $X^{\top}X$ costs $O(np^2)$; the factorization then costs $O(p^3)$ per sweep, with $O(p^2)$ storage for the precision matrix. The changing scales require updating its diagonal and refactorizing. The factorization can also supply a log determinant if needed: $\log |A| = 2 \sum_j \log L_{jj}$.

8.2 An exact observation-space draw when p is large

Bhattacharya, Chakraborty, and Mallick [4] give an exact Gaussian sampling method that can exploit $n \ll p$. Here is its form for our parameterization. Condition on α, σ^2, D and put $H = I_n + XDX^{\top}$.

1. Draw independently $u \sim \mathcal{N}_p(0, D)$ and $\delta \sim \mathcal{N}_n(0, I_n)$.
2. Form $v = Xu + \delta$.
3. Solve $Hw = \tilde{y}/\sigma - v$.
4. Return $\beta = \sigma(u + DX^{\top}w)$.

All normal variables in this construction are freshly drawn each sweep. With diagonal D , generate $u_j = \sqrt{D_{jj}}z_j$ using independent standard normals. Multiplication by D is a vector scaling operation; there is no reason to construct a dense diagonal matrix.

To see why the draw is correct, let $K = DX^{\top}H^{-1}$. Its mean is $K\tilde{y}$, and the Woodbury identity gives

$$A^{-1} = D - DX^{\top}H^{-1}XD, \quad A^{-1}X^{\top} = DX^{\top}H^{-1} = K. \quad (8.2)$$

Also $\text{Cov}(u, v) = DX^{\top}$ and $\text{Var}(v) = H$. Consequently,

$$\text{Var}(u - Kv) = D - KXD - DX^{\top}K^{\top} + KHK^{\top} = D - DX^{\top}H^{-1}XD = A^{-1}.$$

Multiplication by σ gives precisely the mean and covariance in (6.3). This is a different way to sample the same conditional, not an approximation to it.

For a dense design, forming $XDX^{\top}$ costs $O(n^2p)$ and its factorization costs $O(n^3)$. Storage is $O(np + n^2)$, including the design matrix. Thus “linear in p ” assumes n is fixed; it is not linear in both dimensions. For very large scales, cancellation in the final update can still cause numerical trouble, so monitor solver accuracy.

8.3 A coordinatewise alternative

For completeness, each coefficient can also be updated alone. Let $r_{-j} = y - \alpha \mathbf{1}_n - \sum_{k \neq j} X_k \beta_k$ and $a_j = X_j^{\top}X_j + 1/(\tau^2 \lambda_j^2)$. Then

$$\beta_j \mid \beta_{-j}, \cdot \sim \mathcal{N}\left(\frac{X_j^{\top}r_{-j}}{a_j}, \frac{\sigma^2}{a_j}\right). \quad (8.3)$$

A sequential coordinate sweep is valid and can use inexpensive residual updates. With correlated predictors, however, the chain may move slowly along directions that require coordinated changes in several coefficients. A block draw often improves mixing even when it has a larger cost per sweep.

8.4 Numerical details that affect correctness

- **Use the correct gamma convention.** To draw $Z \sim \text{IG}(a, b)$, one option is $G \sim \text{Gamma}(a, \text{rate} = 1)$, then $Z = b/G$. This avoids ambiguities in a library's second parameter. Finite-precision underflow or overflow still requires explicit detection.
- **Record scales on the log scale.** The horseshoe can explore very small and very large positive scales. Log-scale traces expose this behavior more clearly. Stable evaluation of sums or products may be needed before forming reciprocals.
- **Do not silently clip.** Truncating τ , λ_j , or their reciprocals at arbitrary bounds changes the transition and generally the target. If a truncated prior is intended, specify it and derive the corresponding sampler.
- **Distinguish a numerical failure from an ill-defined model.** At finite positive scales, A is positive definite mathematically. A failed Cholesky factorization can indicate overflow, lost symmetry, severe conditioning, or a programming error. Unreported diagonal jitter is an unreported change to the draw.
- **Keep the linear algebra auditable.** Reuse fixed cross-products, check relative solve residuals, and record which Gaussian sampler is used. Avoid explicit inverses and avoid rounding coefficients during the chain.

9 Choosing prior scales and checking their meaning

9.1 Calibrating global shrinkage to anticipated sparsity

Let p_0 be a plausible number of substantial effects, with $0 < p_0 < p$. Under an orthogonal design with common squared column norm c , define the effective number of unshrunk coordinates as

$$m_{\text{eff}} = \sum_{j=1}^p (1 - \kappa_j). \quad (9.1)$$

This is a continuous summary in $(0, p)$, not an integer model size. The sparsity calibration idea follows Piironen and Vehtari [5]; the algebra below uses our noise-scaled convention.

At fixed τ , put $t = \sqrt{c}\tau$. Since $\lambda \sim C^+(0, 1)$,

$$\mathbb{E}(\kappa \mid \tau) = \frac{2}{\pi} \int_0^\infty \frac{d\lambda}{(1 + \lambda^2)(1 + t^2\lambda^2)} = \frac{1}{1 + t}, \quad (9.2)$$

$$\mathbb{E}(m_{\text{eff}} \mid \tau) = p \frac{\sqrt{c}\tau}{1 + \sqrt{c}\tau}. \quad (9.3)$$

Solving $\mathbb{E}(m_{\text{eff}} \mid \tau_0) = p_0$ yields

$$\tau_0 = \frac{p_0}{p - p_0} \frac{1}{\sqrt{c}}. \quad (9.4)$$

For columns with $X_j^\top X_j = n$, a useful calibration is

$$s_\tau = \frac{p_0}{p - p_0} \frac{1}{\sqrt{n}}, \quad \tau \sim C^+(0, s_\tau). \quad (9.5)$$

For example, $n = 100, p = 50, p_0 = 5$ gives $s_\tau = 1/90$. This is an illustrative choice of prior scale, not an empirical estimate.

Equation (9.4) equates an expectation with *fixed* $\tau = \tau_0$. Giving τ a half-Cauchy prior with scale τ_0 does not make the fully averaged prior mean of m_{eff} equal to p_0 . It sets a characteristic global scale (the half-Cauchy median), while retaining considerable uncertainty. Correlated designs and unequal column norms also make this a calibration heuristic rather than an exact model-size statement.

9.2 Why some references include a factor of σ

An alternative writes $\beta_j \mid \tau_{\text{abs}}, \lambda_j \sim \mathcal{N}(0, \tau_{\text{abs}}^2 \lambda_j^2)$, with no explicit σ^2 in the prior variance. Then the orthogonal shrinkage factor is

$$\kappa_j = \frac{1}{1 + c_j \tau_{\text{abs}}^2 \lambda_j^2 / \sigma^2},$$

and the analogous reference scale is $\tau_{\text{abs},0} = \sigma p_0 / \{(p - p_0)\sqrt{n}\}$. In our notation $\tau_{\text{abs}} = \sigma\tau$, so that factor of σ is already accounted for. Multiplying (9.5) by σ again would double-count it. An independent prior on τ_{abs} is also a different joint model from a prior on τ independent of σ .

9.3 The noise-variance prior

The main derivation uses positive a_σ, b_σ . This makes the variance prior proper and keeps its units explicit: b_σ has units of response squared. If $a_\sigma > 1$, its mean is $b_\sigma / (a_\sigma - 1)$; if $a_\sigma > 2$, its variance is $b_\sigma^2 / \{(a_\sigma - 1)^2(a_\sigma - 2)\}$.

Choose parameters using plausible noise levels on the analysis scale, and check the implied prior distribution. For example, $\text{IG}(2, s_0^2)$ has mean s_0^2 but infinite variance; it is a specific heavy-tailed prior, not an automatically innocuous default. Very small inverse-gamma shape and scale values should not be called “noninformative” without examining their implications.

The common scale-invariant prior $p(\sigma^2) \propto 1/\sigma^2$ is improper. Its conditional formula is obtained formally by setting $a_\sigma = b_\sigma = 0$, but this is not a proper $\text{IG}(0, 0)$ distribution. Properness of the resulting posterior must be established for the actual model and data; valid-looking conditional distributions alone do not prove it. A half-Cauchy prior on σ is another conjugately augmentable option, derived in Appendix B.

9.4 Prior predictive reasoning and sensitivity

The prior predicts a regression surface $\alpha \mathbf{1}_n + X\beta$ and responses around it. Ask whether the implied effect sizes and response ranges are scientifically plausible. A literal prior predictive distribution requires a proper intercept prior or a fixed baseline; the flat intercept prior in the main derivation cannot itself generate a proper prior predictive distribution.

For a future fitted analysis, vary p_0 or s_τ , the noise prior, and scientifically relevant predictor scaling choices. Compare substantive coefficient summaries, selected sets, and predictive performance. Sensitivity is especially important when the data weakly identify the global scale or when many predictors are strongly correlated.

10 What to compute from posterior draws

Let $s = 1, \dots, S$ index retained draws from adequately explored chains. For a posterior functional g , a Monte Carlo approximation is

$$\mathbb{E}[g(\theta) \mid y, X] \approx \frac{1}{S} \sum_{s=1}^S g(\theta^{(s)}), \quad (10.1)$$

provided the relevant expectation exists. Dependence between draws affects Monte Carlo precision, which is measured through effective sample sizes and Monte Carlo standard errors rather than through S alone.

10.1 Coefficient summaries and uncertainty

Useful summaries include posterior medians, central credible intervals, probabilities of a positive effect, and probabilities of exceeding a scientifically meaningful magnitude. A posterior mean is useful under squared error loss when it exists. In weakly identified directions under heavy-tailed priors, moment existence and numerical stability should not be assumed from the fact that a finite Monte Carlo average can be calculated.

A central $100(1 - \gamma)\%$ interval for β_j is the interval between its empirical $\gamma/2$ and $1 - \gamma/2$ posterior quantiles. It describes uncertainty conditional on the model, data, and prior. It is not automatically a simultaneous statement about all p coefficients, and does not automatically have nominal repeated-sampling coverage.

When using the orthogonal interpretation, summarize $\kappa_j^{(s)} = 1/\{1 + c_j \tau^{2(s)} \lambda_j^{2(s)}\}$ and $m_{\text{eff}}^{(s)} = \sum_j (1 - \kappa_j^{(s)})$. In general,

$$\mathbb{E} \left[\frac{1}{1 + c_j \tau^2 \lambda_j^2} \mid y \right] \neq \frac{1}{1 + c_j \mathbb{E}(\tau^2 \mid y) \mathbb{E}(\lambda_j^2 \mid y)}.$$

The right-hand side also relies on moments that may be unstable or absent. Compute the nonlinear quantity within each draw and then summarize it.

10.2 Posterior prediction

Transform a new predictor vector with the recorded training-data constants. For every posterior draw, compute

$$\mu_*^{(s)} = \alpha^{(s)} + x_*^\top \beta^{(s)}, \quad y_*^{(s)} = \mu_*^{(s)} + \sigma^{(s)} z_*^{(s)}, \quad z_*^{(s)} \sim \mathcal{N}(0, 1). \quad (10.2)$$

The distribution of $\mu_*^{(s)}$ describes uncertainty about the conditional mean. The distribution of $y_*^{(s)}$ additionally includes observation noise. For multiple new observations, use the same parameter draw for the whole vector and independent noise draws conditional on that state; parameter uncertainty then induces dependence between their predictive distributions.

When the relevant moments exist, total variance gives

$$\text{Var}(y_* \mid y, X) = \mathbb{E}(\sigma^2 \mid y, X) + \text{Var}(\alpha + x_*^\top \beta \mid y, X). \quad (10.3)$$

This identity explains why plugging in coefficients and adding only a noise interval understates predictive uncertainty.

10.3 A general-design measure of conditional complexity

For the penalized slopes, conditional fitted values use the matrix $S_D = X(X^\top X + D^{-1})^{-1}X^\top$. Its trace

$$d_{\text{eff}}(D) = \text{tr}(S_D) \quad (10.4)$$

is a conditional smoothing-complexity summary. In the orthogonal case it is $\sum_j (1 - \kappa_j)$, but in a correlated design the trace accounts for joint directions in predictor space. It is not an exact count of selected variables and is not by itself the degrees of freedom of the fully adaptive posterior procedure. For centered predictors, an unpenalized estimated intercept contributes an additional one to the conditional smoother trace.

11 Variable selection is a posterior decision

11.1 Why there is no ordinary inclusion probability

With a proper posterior under the continuous horseshoe prior,

$$\mathbb{P}(\beta_j = 0 \mid y, X) = 0, \quad \mathbb{P}(\beta_j \neq 0 \mid y, X) = 1. \quad (11.1)$$

Neither probability ranks variables. A spike-and-slab model instead places a point mass at zero and can define a latent inclusion indicator. The ordinary horseshoe has no such indicator. Its proximity to sparse estimates does not turn $1 - \kappa_j$ into a posterior inclusion probability.

A selection report should therefore state the event being judged, the threshold or loss, and whether the goal is scientific interpretation or a smaller predictive model.

11.2 Rule 1: exclusion of zero from a credible interval

One simple rule retains j if a chosen central credible interval for β_j excludes zero. Equivalently, for a continuous posterior, a central $100(1 - \gamma)\%$ interval excludes zero when

$$\mathbb{P}(\beta_j > 0 \mid y) > 1 - \gamma/2 \quad \text{or} \quad \mathbb{P}(\beta_j < 0 \mid y) > 1 - \gamma/2.$$

This rule rewards sign certainty. It can omit important but highly correlated predictors whose individual effects are uncertain. It also does not control the familywise error rate or the frequentist false discovery rate merely because each interval has a familiar nominal level.

11.3 Rule 2: practical relevance and an explicit loss

Choose a threshold $\delta_j > 0$ in meaningful coefficient units and define

$$q_j = \mathbb{P}(|\beta_j| > \delta_j \mid y, X) \approx \frac{1}{S} \sum_s \mathbf{1}\{|\beta_j^{(s)}| > \delta_j\}. \quad (11.2)$$

Unlike exact nonzero probability, this is informative under a continuous prior. For standardized predictors, δ_j can represent a minimum response change for a one-standard-deviation predictor change.

Let $d_j = 1$ mean selecting predictor j . Suppose a false selection costs $C_{\text{FP}} > 0$, and missing a practically relevant effect costs $C_{\text{FN}} > 0$. Under additive loss, the two posterior expected losses are

$$L(d_j = 1 \mid y) = C_{\text{FP}}(1 - q_j), \quad L(d_j = 0 \mid y) = C_{\text{FN}}q_j.$$

Select if and only if

$$q_j > \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}}, \quad (11.3)$$

with a declared convention at equality. This threshold follows from the specified loss, rather than from an arbitrary label of “included.”

For a nonempty reported set S_0 , the posterior expected fraction of practically negligible selected effects is

$$\mathbb{E} \left[\frac{\sum_{j \in S_0} \mathbf{1}\{|\beta_j| \leq \delta_j\}}{|S_0|} \mid y, X \right] = \frac{1}{|S_0|} \sum_{j \in S_0} (1 - q_j). \quad (11.4)$$

One may sort q_j ’s and retain a largest prefix whose average $1 - q_j$ does not exceed a chosen tolerance. Linearity of expectation makes this formula valid without posterior independence. It is a Bayesian statement about practical-null events under the model, not an automatic frequentist false discovery guarantee.

11.4 Rule 3: a shrinkage-score threshold

An exploratory rule ranks variables by $r_j = \mathbb{E}(1 - \kappa_j \mid y, X)$, possibly selecting those above a stated cutoff. This is easy to compute but depends on column scaling and on the orthogonal interpretation. A cutoff such as $1/2$ is a convention unless justified by an explicit decision problem. Report it as a shrinkage-score rule, not as a posterior probability that the variable belongs to a model.

11.5 A smaller predictive model and the effect of correlation

If the goal is a compact predictor, a useful target is to approximate the full model’s fitted or predictive distribution using fewer variables. For example, for a candidate subset S_0 and each posterior draw, one can minimize

$$\min_{a, b_{S_0}} \frac{1}{n} \left\| \alpha^{(s)} \mathbf{1}_n + X \beta^{(s)} - a \mathbf{1}_n - X_{S_0} b_{S_0} \right\|^2.$$

This defines an empirical squared-error projection of the full fitted mean; it is not a complete predictive selection procedure by itself. Choosing a subset size still requires predictive assessment, and uncertainty or residual variance for the reduced predictor must account for approximation error. Perform subset search and tuning within training folds before evaluating on held-out data.

In a correlated group, retaining one representative may preserve predictions even when the data cannot distinguish which member has an individual effect. For scientific interpretation, report that ambiguity, consider prespecified group contributions, and avoid interpreting a selected slope as a causal effect without additional assumptions.

What a transparent selection result should contain

State the preprocessing, prior, decision rule, interval level or practical threshold, selected set, and uncertainty summaries. Include sensitivity to reasonable thresholds and explain how correlated predictors were handled. If a reduced model is refitted, describe its new inferential target rather than treating the refit as the original horseshoe posterior.

12 Diagnostics and a validation plan

12.1 Convergence and Monte Carlo accuracy

Use several independently seeded chains with dispersed reasonable starts. Inspect traces and between-chain agreement for coefficients, $\log \tau$, $\log \sigma$, representative local scales, fitted values, and important decisions. A coefficient near zero can look stable while its global and local scales remain highly dependent and explore slowly.

Rank-normalized split $\hat{R}$, bulk effective sample size, and tail effective sample size are useful diagnostics for heavy-tailed posteriors [6]. Values of $\hat{R}$ close to one, often with a working target below 1.01, are screening criteria rather than proofs of convergence. Assess effective sample sizes and Monte Carlo errors for the actual reported quantities, including interval endpoints and selection-event probabilities. A threshold decision is unreliable if its Monte Carlo error is comparable to its distance from the threshold.

Discard an initial transient based on exploration and diagnostics, and extend runs when necessary. Routine thinning discards information; it is mainly a storage choice, not a remedy for slow mixing. Prefer improving parameterization or block updates and keeping enough draws for accurate summaries. There is no single iteration count that guarantees adequacy for every design and signal configuration.

12.2 Posterior predictive checks

From retained states generate replicated observations $y_i^{\text{rep},(s)} \sim \mathcal{N}(\alpha^{(s)} + x_i^T \beta^{(s)}, \sigma^{2(s)})$. Compare observed and replicated summaries that address the likelihood: residual spread versus fitted value, extreme residuals, skewness, and dependence across time or groups where relevant. A well-mixed chain can still fit a misspecified homoscedastic Gaussian model. Predictive checking and sampler convergence answer different questions.

12.3 Checks for a future implementation

The following are proposed checks, not completed experiments in these notes:

1. **Distribution convention.** At fixed a, b , confirm that the inverse-gamma generator agrees with the reciprocal-gamma relationship, using quantiles rather than nonexistent moments when $a \leq 1$.
2. **Gaussian block at fixed scales.** Compare sampled means and covariances against the exact normal target for a small problem. Use a non-diagonal precision so a transposed-Cholesky bug is detectable. Compare both Gaussian algorithms within Monte Carlo error.
3. **Orthogonal reduction.** Check that the block sampler reproduces (3.2) for orthogonal columns, with the correct column norm. Include $c_j = n$, not only $c_j = 1$.
4. **Conditional log densities.** Vary one block in (5.1) while holding the others fixed and verify agreement, up to an additive constant, with its proposed full-conditional log density. This catches omitted normalizing factors.

5. **Intercept and rank cases.** Compare the explicit and correctly integrated intercept formulations; include correlated, duplicated, and $p > n$ designs. Check both predictions and noise-variance behavior.
6. **End-to-end calibration.** With proper priors for every parameter, including the intercept, consider simulation-based calibration and comparisons with a separately implemented posterior method. Such tests must account for Monte Carlo dependence and uncertainty.

12.4 Reproducibility for future computational work

Any future simulations, benchmarks, or computational validation for this project should run as persistent jobs on `owls.stat.rice.edu`, after reading its applicable `AGENTS.md` instructions. Local editing and document compilation do not constitute a sampler experiment. These notes contain no claim that a numerical sampler has been run.

A computational run should preserve the exact source and configuration, data/preprocessing description, random seeds and generator details, SHA-256 hashes, launch command, scheduler or process identifiers, environment and package versions, timestamps, exit status, logs, generated data, retained draws, and diagnostic evidence. Conclusions should be based on inspected saved outputs. If the remote system is unavailable, report that fact instead of substituting a local experiment.

A Intercept integration and collapsed updates

A.1 Integrating out the flat intercept exactly

Assume $n \geq 2$, $\mathbf{1}_n^\top X = 0$, and set $y_c = y - \bar{y}\mathbf{1}_n$. The residual identity is

$$\|y - \alpha\mathbf{1}_n - X\beta\|^2 = \|y_c - X\beta\|^2 + n(\alpha - \bar{y})^2.$$

Integrating the likelihood over α introduces the factor $\sqrt{2\pi\sigma^2/n}$. Therefore the integrated likelihood is

$$p(y | \beta, \sigma^2) \propto (\sigma^2)^{-(n-1)/2} \exp \left\{ -\frac{\|y_c - X\beta\|^2}{2\sigma^2} \right\}, \quad (\text{A.1})$$

where the proportionality reflects the arbitrary constant in the flat intercept prior. The coefficient update uses $\tilde{y} = y_c$, while the variance update is

$$\sigma^2 | \beta, D, y \sim \text{IG} \left(a_\sigma + \frac{n-1+p}{2}, b_\sigma + \frac{\|y_c - X\beta\|^2 + \beta^\top D^{-1}\beta}{2} \right). \quad (\text{A.2})$$

Other scale updates are unchanged. To recover intercept uncertainty, draw $\alpha \sim \mathcal{N}(\bar{y}, \sigma^2/n)$ conditional on retained states. For uncentered predictors the analogous formula uses the centering projector $M = I_n - \mathbf{1}_n\mathbf{1}_n^\top/n$, so replace X and y by MX and My .

Subtracting $\bar{y}$ from the response but using an n -dimensional zero-intercept likelihood with the original $n/2$ power of σ^2 would instead treat the estimated baseline as known. That is a different variance calculation.

A.2 Integrating out the coefficients for a joint block

Hold α, D fixed and integrate out β . Normal convolution gives

$$\tilde{y} | \sigma^2, D \sim \mathcal{N}_n(0, \sigma^2 H), \quad H = I_n + XDX^\top. \quad (\text{A.3})$$

Therefore

$$\sigma^2 | \alpha, D, y \sim \text{IG} \left(a_\sigma + \frac{n}{2}, b_\sigma + \frac{\tilde{y}^\top H^{-1} \tilde{y}}{2} \right). \quad (\text{A.4})$$

Here there is no $p/2$: the coefficients have actually been integrated out, including their normalizing factors. This explains how two variance updates with different shapes can both be correct for different conditionals.

A valid joint-block replacement for the main sweep is:

1. update α conditional on the current β, σ^2 ;
2. draw σ^2 using (A.4), conditional on the current α, D ;
3. immediately draw β using (6.3) with that new σ^2 , before updating any scale conditioned on β ;
4. update the local, global, and auxiliary scales as before.

Steps 2–3 together draw from the joint conditional of (σ^2, β) . Inserting a collapsed variance formula somewhere inside an otherwise unchanged scan without respecting the relevant block ordering can produce an invalid chain.

A.3 Properness under the main variance prior

With $n \geq 1$, $a_\sigma, b_\sigma > 0$, proper scale priors, and the flat intercept prior, the posterior in the main model is proper. One way to see this is to integrate α first. Its integrated likelihood is at most a constant times $(\sigma^2)^{-(n-1)/2}$, since the residual exponential is at most one. Integrating the proper coefficient and scale priors cannot increase that bound. Finally,

$$\int_0^\infty (\sigma^2)^{-(n-1)/2} p(\sigma^2) d\sigma^2 < \infty$$

under a proper inverse-gamma prior with positive scale. The likelihood is positive for finite data, so the normalizing constant is also positive. This argument does not carry over automatically to $p(\sigma^2) \propto 1/\sigma^2$, and properness alone does not ensure existence of every posterior moment.

B Alternative prior choices change specific updates

B.1 A coefficient prior independent of the noise variance

Consider the explicitly different hierarchy

$$\beta \mid \tau_{\text{abs}}^2, \lambda^2 \sim \mathcal{N}_p(0, D_{\text{abs}}), \quad D_{\text{abs}} = \tau_{\text{abs}}^2 \text{diag}(\lambda_j^2),$$

with $\tau_{\text{abs}} \sim \mathcal{C}^+(0, s_{\text{abs}})$ independent of σ . The likelihood, flat intercept prior, and proper inverse-gamma noise prior remain as in the main model. Using the analogous augmentation, the updated conditionals are

$$\begin{aligned} V &= (X^\top X / \sigma^2 + D_{\text{abs}}^{-1})^{-1}, \\ \beta \mid \cdot &\sim \mathcal{N}_p(V X^\top \tilde{y} / \sigma^2, V), \end{aligned} \tag{B.1}$$

$$\sigma^2 \mid \cdot \sim \text{IG}\left(a_\sigma + \frac{n}{2}, b_\sigma + \frac{R}{2}\right), \tag{B.2}$$

$$\lambda_j^2 \mid \cdot \sim \text{IG}\left(1, \frac{1}{\nu_j} + \frac{\beta_j^2}{2\tau_{\text{abs}}^2}\right), \tag{B.3}$$

$$\tau_{\text{abs}}^2 \mid \cdot \sim \text{IG}\left(\frac{p+1}{2}, \frac{1}{\xi} + \frac{1}{2} \sum_j \frac{\beta_j^2}{\lambda_j^2}\right), \tag{B.4}$$

$$\nu_j \mid \cdot \sim \text{IG}(1, 1 + 1/\lambda_j^2), \tag{B.5}$$

$$\xi \mid \cdot \sim \text{IG}(1, 1/s_{\text{abs}}^2 + 1/\tau_{\text{abs}}^2). \tag{B.6}$$

The intercept update is unchanged. Relative to the main model, the noise update loses both the coefficient-prior penalty and the $p/2$ shape contribution; the local and global updates lose their σ^2 factors. One cannot mix this variance update with the main model's coefficient update.

B.2 A half-Cauchy prior on the noise standard deviation

Return to the main, noise-scaled coefficient prior, but replace (2.6)'s variance prior by

$$\sigma \sim \mathcal{C}^+(0, s_\sigma), \quad \sigma^2 \mid \eta \sim \text{IG}(1/2, 1/\eta), \quad \eta \sim \text{IG}(1/2, 1/s_\sigma^2).$$

The resulting updates are

$$\sigma^2 \mid \cdot \sim \text{IG} \left(\frac{n+p+1}{2}, \frac{1}{\eta} + \frac{R+Q}{2} \right), \quad (\text{B.7})$$

$$\eta \mid \cdot \sim \text{IG} \left(1, \frac{1}{s_\sigma^2} + \frac{1}{\sigma^2} \right). \quad (\text{B.8})$$

All other main-model conditionals are unchanged. These formulas condition on the explicit intercept. The appropriate residual-dimension change is needed if the intercept is integrated out. Because the variance prior now behaves differently near zero, the inverse-gamma properness bound in Appendix A should not be applied without checking its assumptions.

B.3 Fixing the global scale

If $\tau = \tau_0 > 0$ is fixed by the model, omit both the τ^2 and ξ updates. Keep the remaining conditionals with that fixed value. This is a useful mathematically simpler model, but it excludes posterior uncertainty about global shrinkage. Choosing τ_0 by an optimization or empirical-Bayes step is a separate procedure and should be reported.

B.4 The regularized horseshoe

The regularized horseshoe adds control over the largest coefficient scales [5]. In a noise-scaled convention, one possible form is

$$\tilde{\lambda}_j^2 = \frac{c^2 \lambda_j^2}{c^2 + \tau^2 \lambda_j^2}, \quad \beta_j \mid \cdot \sim \mathcal{N}(0, \sigma^2 \tau^2 \tilde{\lambda}_j^2), \quad c > 0. \quad (\text{B.9})$$

For $\tau^2 \lambda_j^2 \ll c^2$, this approximately recovers the ordinary horseshoe variance. For $\tau^2 \lambda_j^2 \gg c^2$, the variance tends to $\sigma^2 c^2$, limiting the scale of very large effects. Here c is measured relative to the noise standard deviation; a slab width in absolute response units requires a corresponding reparameterization.

This modification changes the prior normalization as a function of λ_j^2 and τ^2 . The simple inverse-gamma local and global updates derived for the ordinary horseshoe no longer apply unchanged. Re-derive the target and use an appropriate sampler; substituting $\tilde{\lambda}_j^2$ in one line of the ordinary Gibbs sweep is not a valid derivation.

C A one-predictor conditional example

To make each quantity concrete, suppose one centered predictor has $X^\top X = 20$, and the currently intercept-adjusted response satisfies $X^\top \tilde{y} = 10$ and $\tilde{y}^\top \tilde{y} = 15$. Take a current state $\sigma^2 = 2, \tau^2 = 0.25, \lambda^2 = 4$. Then $D = 1, A = 21$, and the coefficient update is

$$\beta \mid \cdot \sim \mathcal{N}(10/21, 2/21).$$

The unregularized coordinate estimate is $10/20 = 0.5$; its conditional shrinkage factor is $\kappa = 1/21$, so its conditional posterior mean is $(20/21)0.5 = 10/21$, agreeing with the matrix calculation.

For illustration of the next conditional, evaluate the formulas at $\beta = 0.4$, with $n = 20, p = 1, a_\sigma = 2, b_\sigma = 1$. The residual sum of squares and prior penalty are

$$R = 15 - 2(0.4)(10) + (0.4)^2(20) = 10.2, \quad Q = (0.4)^2 = 0.16.$$

Thus

$$\sigma^2 \mid \cdot \sim \text{IG}(12.5, 6.18).$$

If evaluating the local-scale conditional at $\nu = 1$ and $\sigma^2 = 2$, while holding $\tau^2 = 0.25$, then

$$\lambda^2 \mid \cdot \sim \text{IG}\left(1, 1 + \frac{0.16}{2(2)(0.25)}\right) = \text{IG}(1, 1.16).$$

These are separate conditional evaluations at stated values, not successive realized Gibbs draws. In an actual sweep, the local-scale update must use the newly drawn noise variance. No random experiment is being reported here.

D Quick reference for the main sampler

Model. Explicit flat-prior intercept; noise-scaled coefficient prior; $\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma)$; $\lambda_j \sim \text{C}^+(0, 1)$; $\tau \sim \text{C}^+(0, s_\tau)$. The inverse-gamma density is proportional to $z^{-a-1}e^{-b/z}$.

Definitions.

$$\begin{aligned} D &= \tau^2 \text{diag}(\lambda_1^2, \dots, \lambda_p^2), & A &= X^\top X + D^{-1}, \\ \tilde{y} &= y - \alpha \mathbf{1}_n, & m &= A^{-1} X^\top \tilde{y}, \\ R &= \|\tilde{y} - X\beta\|^2, & Q &= \sum_j \frac{\beta_j^2}{\tau^2 \lambda_j^2}. \end{aligned}$$

Full conditionals. In a sequential scan, always use current values.

$$\begin{aligned} \alpha \mid \cdot &\sim \mathcal{N}\left(\frac{\mathbf{1}_n^\top (y - X\beta)}{n}, \frac{\sigma^2}{n}\right), \\ \beta \mid \cdot &\sim \mathcal{N}_p(m, \sigma^2 A^{-1}), \\ \sigma^2 \mid \cdot &\sim \text{IG}\left(a_\sigma + \frac{n+p}{2}, b_\sigma + \frac{R+Q}{2}\right), \\ \lambda_j^2 \mid \cdot &\sim \text{IG}\left(1, \frac{1}{\nu_j} + \frac{\beta_j^2}{2\sigma^2 \tau^2}\right), \\ \tau^2 \mid \cdot &\sim \text{IG}\left(\frac{p+1}{2}, \frac{1}{\xi} + \frac{1}{2\sigma^2} \sum_j \frac{\beta_j^2}{\lambda_j^2}\right), \\ \nu_j \mid \cdot &\sim \text{IG}\left(1, 1 + \frac{1}{\lambda_j^2}\right), \\ \xi \mid \cdot &\sim \text{IG}\left(1, \frac{1}{s_\tau^2} + \frac{1}{\tau^2}\right). \end{aligned}$$

Interpretation. Small τ encourages global shrinkage; large λ_j allows a local exception. An orthogonal-design shrinkage factor is $\kappa_j = 1/\{1 + (X_j^\top X_j)\tau^2 \lambda_j^2\}$. The horseshoe is continuous; exact-zero selection requires a separate decision rule or a different prior.

Most consequential mistakes. Confusing gamma rate and scale; omitting the $p/2$ or Q term in the main noise update; shrinking the intercept unintentionally; forgetting column norms in κ_j ; using the wrong Cholesky transpose; replacing random draws by means; simultaneously updating dependent blocks with stale states; and reporting a shrinkage score as an inclusion probability.

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